ModelIt!

Teacher's Guide

# Can Cities Run on 100% Clean Energy?

Modeling Renewable Grid Stability, Storage, and Intermittency

## Quick Reference

| | |
|---|---|
| **Grade Level** | Grade 11L2 |
| **NGSS Standard** | HS-PS3-3, HS-ESS3-2 |
| **Time** | 45–60 minutes (core lesson) |
| **Materials** | Student Activity Pack, devices with ModelIt! access |
| **Prep** | Print activity packs, test ModelIt! on student devices |
| **Grouping** | Partners or small groups (2–4) |

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# 1 NGSS Standards & CAST Alignment

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**Performance Expectation: HS-PS3-3, HS-ESS3-2**

Design, build, and refine a device that works within given constraints to convert one form of energy into another form of energy. Evaluate competing design solutions for developing, managing, and utilizing energy and mineral resources based on cost-benefit ratios.

## Three-Dimensional Learning

| | |
|---|---|
| Science Practice | **Designing Solutions and Using Computational Thinking** — Students design energy system solutions using computational models to optimize the balance between renewable generation, storage capacity, and grid reliability under variable weather conditions. |
| Disciplinary Core Idea | **PS3.D Energy in Chemical Processes / ESS3.A Natural Resources** — Students apply energy conversion principles to understand how electricity is generated from renewable sources and stored in chemical or mechanical form, while evaluating the resource requirements of different storage technologies. |
| Crosscutting Concept | **Systems and System Models / Energy and Matter** — Students model the energy grid as a complex system where energy flows from generation through storage to consumption, analyzing how intermittent inputs and variable demand create system-level challenges. |

## CAST Testing Alignment

**MULTIPLE CHOICE:**

A city at 90% renewable energy experiences a week of overcast skies and calm winds in January. Battery storage is depleted after 36 hours. Based on your model, which outcome is MOST likely?

**CONSTRUCTED RESPONSE:**

Using your model, explain why the cost of achieving 100% renewable electricity is not simply double the cost of achieving 50%. Reference the duck curve, seasonal variation, and storage requirements in your explanation.

# 2 Scoring Rubric

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### Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

| Skill | Expert (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|
| **Identifying Components** | EXPERT<br>Correctly identifies and sorts ALL components as internal or external with detailed reasoning | PROFICIENT<br>Correctly identifies and sorts MOST components with some reasoning | DEVELOPING<br>Identifies components but has difficulty sorting or explaining | BEGINNING<br>Needs significant help identifying components |
| **Establishing Relationships** | EXPERT<br>Accurately describes ALL positive and negative relationships with clear evidence | PROFICIENT<br>Accurately describes MOST relationships with some evidence | DEVELOPING<br>Identifies some relationships but struggles with direction | BEGINNING<br>Needs help understanding how components connect |
| **Running Simulations** | EXPERT<br>Makes accurate predictions AND explains surprising results using evidence | PROFICIENT<br>Makes reasonable predictions and records clear observations | DEVELOPING<br>Runs simulations but predictions or observations are incomplete | BEGINNING<br>Needs help running simulations or recording data |
| **Explaining Results** | EXPERT<br>Uses multiple pieces of evidence to explain HOW and WHY systems interact | PROFICIENT<br>Uses evidence to explain system interactions | DEVELOPING<br>Gives partial explanations with limited evidence | BEGINNING<br>Needs help connecting evidence to explanations |

### Scoring Guidance

**Expert (4):** Student demonstrates deep understanding and can transfer knowledge to new contexts.
**Proficient (3):** Student meets grade-level expectations and shows solid understanding.
**Developing (2):** Student is making progress but needs additional scaffolding or practice.
**Beginning (1):** Student needs significant support; consider using Differentiation strategies (Section 7).

# 3 Background Content Knowledge

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**For Your Confidence**

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

The transition to 100% renewable electricity is the defining engineering challenge of the 21st century. While solar and wind costs have dropped 90% in the past decade, making them the cheapest sources of new electricity generation, their intermittent nature creates a systems integration challenge that grows exponentially as renewable penetration increases. Understanding this challenge requires modeling energy as a complex system with daily, weekly, and seasonal dynamics.

## The Intermittency Challenge

Solar panels produce electricity only during daylight hours, with output peaking at midday and dropping to zero at night. Cloud cover can reduce output by 80% within minutes. Wind turbines produce power when wind speeds are between 7-55 mph, but calm periods lasting days are common. In most locations, solar output is 2-3 times higher in summer than winter, while heating demand peaks in winter — creating a seasonal mismatch. A grid relying on these sources must somehow match supply to demand every second of every day, even when nature doesn't cooperate.

## The Duck Curve and Daily Cycling

As solar penetration increases, midday electricity production increasingly exceeds demand. California's famous 'duck curve' shows net demand (total demand minus solar) plunging at midday and then surging in the evening as solar drops and people come home. This creates two problems: curtailment (wasting solar energy because the grid can't absorb it) and ramping (needing to bring massive generation online in just 3 hours as solar disappears). In 2023, California curtailed over 2.4 million MWh of solar energy — enough to power 350,000 homes — while still relying on natural gas for evening peaks.

## Storage Technologies and Their Limits

Lithium-ion batteries currently dominate grid storage, with costs falling to $150/kWh for 4-hour systems. They excel at daily cycling — absorbing midday solar surplus and discharging during evening peaks. But they become prohibitively expensive for multi-day or seasonal storage because you need enormous capacity. Pumped hydro storage (pumping water uphill to store energy) provides 93% of global energy storage but requires specific geography. Emerging technologies include compressed air, iron-air batteries (100-hour duration), and green hydrogen (seasonal storage). Each technology has different cost, duration, efficiency, and geographic constraints.

## Real-World Progress and Remaining Gaps

Several regions have achieved high renewable penetration: South Australia regularly exceeds 70% renewable, Denmark hits 80%+ on windy days, and Costa Rica runs on 98% renewable (mostly hydro). But these success stories rely on interconnection with neighbors who provide backup. A truly isolated 100% renewable grid has never been achieved at city scale. Studies suggest that getting from 80% to 100% renewable requires 5-10x more storage per additional percentage point compared to getting from 0% to 50%. The final 10% may require seasonal storage technologies that don't yet exist at commercial scale.

# 4 The LEVER Framework



### About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

**L** **Locate the System**

Students identify six components of the renewable energy grid system: Solar Generation Capacity, Wind Generation Capacity, Energy Storage Capacity, Grid Demand Pattern, Supply-Demand Balance, and Grid Reliability Index — distinguishing controllable inputs from system responses.

**E** **Establish Relationships**

Students determine that Solar and Wind generation create variable supply that must match Grid Demand Pattern through Energy Storage. Supply-Demand Balance determines Grid Reliability. The duck curve emerges when solar peaks misalign with demand peaks.

**V** **Visualize & Model**

Students build a computational model showing hourly energy generation, storage cycling, and demand matching across daily and seasonal timescales.

**E** **Evaluate Outcomes**

Students run 50%, 90%, and 100% renewable scenarios to quantify how storage requirements scale nonlinearly with renewable penetration and identify the reliability limits of each configuration.

**R** **Revise & Extend**

Students add Demand Response Programs, Green Hydrogen, or Geographic Diversification to explore advanced strategies for closing the final gap to 100% renewable electricity.

## Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

| | |
|---|---|
| **At the start of a unit** | Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it. |
| **After a reading or lab** | Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes. |
| **As a “what-if” tool** | When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back. |
| **For peer discussion** | Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence. |
| **As a presentation tool** | Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time. |
| **At the end of a unit** | Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved. |

# 5 Slide-by-Slide Facilitation

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## Slide 1: Cover

2 min

**Say:** *"What if I told you that California threw away enough solar energy last year to power 350,000 homes — while still burning natural gas for electricity? That paradox is what we're modeling today."*

**Watch For**

Show the California duck curve graph. Let students identify the 'duck.' Ask: Why would we waste clean energy while also burning fossil fuels?

## Slide 2: Learning Objectives

3 min

**Say:** *"Today you're tackling one of the biggest engineering challenges of your lifetime — making the electrical grid run on nature's schedule."*

**Watch For**

Have students read objectives. Pre-teach 'intermittency' and 'grid stability.' Quick activity: turn off the classroom lights for 3 seconds. That's what a grid failure feels like — except for a hospital.

## Slide 3: Big Question

3 min

**Say:** *"Solar costs have dropped 90% in ten years. Wind is now the cheapest electricity on Earth. So why aren't we at 100% renewable already? The answer isn't cost — it's physics."*

**Watch For**

Quick-write: What happens at 8 PM on a cloudy, windless winter night? Where does electricity come from? Pair-share responses.

## Slide 4: LEVER Framework

2 min

**Say:** *"We're modeling an entire city's electrical grid — millions of people's power, every second of every day, through every weather pattern."*

**Watch For**

Preview each LEVER step. Emphasize that this is a dynamic system where supply and demand must balance in real time — there's no inventory of electricity.

## Slide 5: Activity 1: Components

8 min

**Say:** *"Three things you can build — solar, wind, and storage. Three things the system determines — demand pattern, balance, and reliability. How do they connect?"*

**Watch For**

Guide sorting of external vs. internal components. Discuss why Solar, Wind, and Storage are external (we choose how much to build) while Demand, Balance, and Reliability are system responses.

## Slide 6: Activity 2: Connections

8 min

**Say:** *"When solar generation peaks at noon but demand peaks at 7 PM, what bridges the gap? And what happens when the bridge isn't big enough?"*

**Watch For**

Help students map the timing mismatches. Draw the duck curve on the board. Identify where storage absorbs surplus and where it discharges. Find the failure point.

## Slide 7: Activity 3: Simulation

10 min

**Say:** *"Let's push the grid to its limits. 50% renewable — easy. 90% — challenging. 100% — let's see if it breaks."*

**Watch For**

Students predict storage requirements for each level before running simulations. Focus on the nonlinear jump in storage needs from 80% to 100%. Compare curtailment vs. shortage events.

## Slide 8: Discoveries

5 min

**Say:** *"Your model just showed why the last 20% is ten times harder than the first 80%. But it also showed it's not impossible — just expensive and complex."*

**Watch For**

Lead discussion on why 100% renewable requires systems thinking, not just more panels. Connect to real-world examples: South Australia, Denmark, California. Discuss the role of emerging storage technologies.

## Slide 9: STEM Challenge

12 min

**Say:** *"You're now city energy planners. Half a million people are counting on you. Design a grid that never goes dark — using only clean energy."*

### Watch For

Groups design comprehensive energy portfolios specifying solar/wind ratios, storage types and capacities, and demand management strategies. Present as city council proposals with cost estimates.

# 6 Answer Key & Expected Responses

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## Component Sorting

- **EXTERNAL:** Solar Generation Capacity, Wind Generation Capacity, Energy Storage Capacity
- **INTERNAL:** Grid Demand Pattern, Supply-Demand Balance, Grid Reliability Index

## Relationships

- **Solar Generation Capacity to Supply-Demand Balance:** POSITIVE (+) — Higher Solar Generation increases daytime electricity supply, improving Supply-Demand Balance during daylight hours. However, this relationship reverses at night when solar output drops to zero, creating the daily cycling challenge.
- **Wind Generation Capacity to Supply-Demand Balance:** POSITIVE (+) — Higher Wind Generation adds electricity supply that partially complements solar — wind often produces more at night and in winter when solar is weakest. This diversification improves overall Supply-Demand Balance across more hours.
- **Energy Storage Capacity to Grid Reliability Index:** POSITIVE (+) — More storage allows the grid to absorb excess renewable generation and discharge during gaps, directly increasing the percentage of time demand is met. The relationship is strongly positive but with diminishing returns — each additional hour of storage covers increasingly rare weather events.
- **Supply-Demand Balance to Grid Reliability Index:** POSITIVE (+) — Better Supply-Demand Balance directly improves Grid Reliability — every hour where supply meets demand is an hour without blackouts. The relationship is binary at each moment: either demand is met or it isn't.

## Key Discoveries

- The first 50-60% of renewable energy on a grid is relatively straightforward to integrate because existing backup plants can fill gaps — but the last 20% from 80% to 100% requires exponentially more storage
- The duck curve creates a paradox: midday solar overproduction must be curtailed (wasted) while evening demand peaks require stored energy — solving this requires massive battery deployment or demand shifting
- Seasonal variation is the hardest challenge: a city might need 4 hours of storage to handle daily solar cycles but 4 weeks of storage to handle winter solar droughts — a 170x difference in storage requirement
- Grid reliability of 99.97% requires that even the worst-case weather scenario (a week of clouds and calm wind in winter) can be covered by storage alone — this sets the minimum storage requirement

# 7 Collaborative Learning Protocols

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### Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

### Model Gallery Walk

**When:** After students build models (LEVER Phase V)
**Setup:** Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
**Protocol:** Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
**Why it works:** Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

### Scientist Fishbowl

**When:** During simulation discussion (LEVER Phase E)
**Setup:** 4-5 students sit in center circle. Rest of class observes with clipboards.
**Protocol:** Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
**Why it works:** Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

### Component Jigsaw

**When:** During component analysis (LEVER Phase L)
**Setup:** Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
**Protocol:** Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
**Why it works:** Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

### Prediction Debate

**When:** Before running simulations (LEVER Phase E)
**Setup:** Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
**Protocol:** Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
**Why it works:** Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

### Pair Programming for Models (Digital Collaboration)

**When:** Throughout model building
**Setup:** Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
**Protocol:** Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
**Why it works:** Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

### Quick Protocol Selection Guide

| | |
|---|---|
| **Short on time (5 min)** | Prediction Debate — gets bodies moving, generates energy fast |
| **Full period available** | Component Jigsaw — deep collaboration, every student contributes |

| | |
|---|---|
| **After simulation day** | Model Gallery Walk — peer feedback, multiple perspectives |
| **Advanced discussion** | Scientist Fishbowl — builds academic discourse skills |
| **Every day default** | Pair Programming — transforms individual work into dialogue |

# 8 Differentiation & Extensions

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### Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

### Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

### English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

### Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

# 9 Common Misconceptions

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### Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

| STUDENTS MAY THINK | SCIENTIFIC UNDERSTANDING |
|---|---|
| "Solar and wind are unreliable so we can't depend on them" | Solar and wind are variable, not unreliable — their output patterns are predictable and well-understood. Solar will always peak at midday and wind follows seasonal patterns. The challenge is engineering the system to handle variability, not the sources themselves. With sufficient storage and geographic diversification, variable sources can provide reliable electricity. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "We just need more solar panels" | Adding more solar panels increases midday production but does nothing for nighttime, cloudy days, or winter months. Beyond a certain point, additional solar increases curtailment (wasted energy) more than it increases useful generation. The bottleneck isn't generation — it's storage and grid integration. |
| **STUDENTS MAY THINK** | **SCIENTIFIC UNDERSTANDING** |
| "Batteries can store unlimited energy" | All storage technologies have physical and economic limits. Lithium-ion batteries are excellent for hours but prohibitively expensive for weeks. A city needing 7 days of backup would require battery capacity costing more than the entire generation infrastructure. Different timescales need different storage technologies, and some (like seasonal storage) remain unsolved at commercial scale. |